

HOMEWORK 9

NUMERICAL ANALYSIS

(MATH 461) FALL, 2023

DUE: 12/01/2023

Problem 1. Given $h > 0$, find A, B, and C in the following rule, which should give exact results for polynomials of degree 2:

$$\int_{-h}^{3h} f(x)dx \approx h(Af(0) + Bf(h) + Cf(2h)).$$

Let $f(x) = e^{-x}$. What is the leading-order error of the above integration rule?

Problem 2. Write programs that compute an integral numerically, using each of the composite trapezoidal, composite midpoint, and composite Simpson methods. Input for the program consists of the integrand, the ends of the integration interval, and the number of (uniform) subintervals. Then, apply your program to the two following integrals:

(a) $\int_0^1 \frac{4}{1+x^2} dx = \pi$

(b) $\int_0^1 \sqrt{x} dx = \frac{2}{3}$

using 2, 4, 8, 16, and 32 subintervals. For each of the integrals, explain the behavior of the error by observing how it is reduced each time the number of subintervals is doubled, and how large it is, compared to analytical bounds on the error. As much as you can, compare the performance of the methods to each other, taking into consideration both the error and the number of function evaluations.

Problem 3. Find nodes x_0, x_1 and coefficients c_0, c_1 such that

$$\int_{-1}^1 f(x)dx \approx c_0 f(x_0) + c_1 f(x_1)$$

has the highest degree of accuracy possible.